

Submission Summary

Conference Name

IEEE Chandigarh Subsection International Conference

Track Name

Artificial Intelligence and Data Science

Paper ID

1461

Paper Title

Weighted Ensemble Steganalysis: Integrating CNN, XGBoost, and LightGBM

Abstract

Digital image steganography poses a significant challenge to digital forensics due to the microscopic nature of hidden data. Existing detection methods typically rely either on handcrafted statistical features (Traditional Steganalysis) or end-to-end deep learning models (Steganalysis CNNs). This paper proposes a novel Weighted Heterogeneous Ensemble Model (WHEM) that synergistically integrates both approaches to maximize detection accuracy and robustness. The WHEM combines the prediction probabilities from:

- 1) A specialized Steganalysis Convolutional Neural Network (CNN) structure, derived from a GAN Discriminator, operating on image residual noise; 2) XGBoost, and 3) LightGBM, both operating on statistical domain features. Crucially, the final prediction is determined by a weight optimization search that maximizes the overall F1-score.

Experimental results on a simulated dataset demonstrate that this heterogeneous ensemble achieves a classification accuracy of $\approx 74.50\%$, a statistically significant improvement over all individual component models. This proves that combining models with complementary views of the data—raw noise vs. statistical patterns—is an effective strategy for robust steganalysis.

Created

2026-05-05 09:16:45 GMT+5:30

Last Modified

2026-05-05 09:16:45 GMT+5:30

Authors**Jatin Khanijo** (Chitkara University) <jatin2583.be22@chitkara.edu.in>**Submission Files**

2210992522-2210991282-2210992583-2210992421_Research_Paper 1.pdf (337 Kb, 2026-05-05 09:07:47 GMT+5:30)

Submission Questions Response

1. Corresponding author Email ID

Provide Email ID of corresponding author

ankur1282.be22@chitkara.edu.in

vibhor2522.be22@chitkara.edu.in

sunny2421.be22@chitkara.edu.in

isha.kansal@chitkara.edu.in

2. Contact number with country code

Provide Whatsapp number of corresponding author with country code

+91 7988468671

+91 9499185016

+91 8240727229

+91 9592842249

3. Author's Information

I confirm that I have listed the title of the paper and the names, email addresses and affiliations of all contributed authors in the submission system that match the order as given in the submitted paper

Agreement accepted

4. Co-author's Consent

I confirm that I have discussed about this submission with all co-authors and authors have approved the paper and agree with its submission to this conference.

Agreement accepted

5. Certificate of Originality

I confirm that neither the paper nor any parts of its content are currently under consideration or published in another conference/journal and the paper is free from plagiarism

Agreement accepted

6. Author's Consent

It is confirmed that at least one author will register in full to attend and present the paper at the Conference.

Agreement accepted

7. The use of artificial intelligence (AI)-generated text (e.g., by ChatGPT)

I confirm that no text in the paper is AI-generated.

Agreement accepted
